

KNIGHT INDUSTRIES · SPACEFOAM PROGRAM

A Two-Lens Electromagnetic Foam Search on Spacecraft Data

*What a magnetometer / field foam-signature pipeline does, and what its inner-heliosphere
and Voyager outputs actually show*

Status: Null — strong engineering and an honest design, but the Voyager candidate counts are
dominated by numerical artifacts, not foam

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Computational analysis was AI-assisted; the framing, interpretation, and the decision to report this as a null are the author's own.

Summary

What was assessed. A well-built, three-program pipeline that searches spacecraft electromagnetic data for two foam-motivated signatures: persistent narrowband spectral lines (a *steady-state lens*), and delayed post-spike “echoes” with a downward frequency shift (a *transient lens*, the Earth Schumann template applied to flyby spikes). The engineering is genuinely strong — constant-memory streaming, resumable runs, a statistically principled colored-noise null in the steady-state lens, and a README that is unusually honest about what a “candidate” is and is not.

What the outputs show. Two result files were provided, both from the transient lens. The inner-heliosphere run (Parker Solar Probe / Solar Orbiter / MESSENGER / Helios) produced **zero events** — a header-only file. The Voyager run produced **117,370 drive spikes, of which 105,996 (90.3%) were flagged candidate=1**. That 90% rate is the first warning: a flag that fires on nine of every ten events is not discriminating anything. On inspection the flags are dominated by two artifacts. **(1)** the echo significance *echo_z* reaches 10^{29} – 10^{36} — impossible as a real statistic — because the robust trailing baseline collapses to ~zero on flat/quantized data and the detector divides by it. **(2)** the “Schumann-like downward frequency shift” is not systematic: the shift distribution is centered on zero (median +0.001 Hz), and the “downshift” tag simply fires on its negative half. There is no support here for the Earth echo result generalizing to spacecraft data; the run is best read as a cautionary null.

1. What the pipeline is

The pipeline is three cooperating programs plus a shared library, all aimed at one question: does spacetime foam leave a measurable fingerprint in spacecraft magnetometer and field data? It is worth separating the engineering (which is strong) from the result (which is null).

download_em_data.py — a CDAWeb downloader for the inner-heliosphere and Mercury missions, with a SQLite manifest, atomic part-files, and HTTP-range resume.

analyze_em_foam.py — the *steady-state lens*. For each window it computes a Welch PSD of $|B|$, fits a robust running-median continuum, and tests for narrowband excess against a **genuinely principled null**: with K averaged Welch segments the PSD-to-continuum ratio follows a $\text{Gamma}(K, 1/K)$ distribution, and the detection threshold is the Bonferroni-corrected inverse survival function of that Gamma. It adds a quality factor $Q = f_0/\Delta f$, a frequency-drift track, and a known-instrument-line veto. This is the right way to find lines against a red turbulent background.

analyze_em_echo.py — the *transient lens*, and the one that produced the provided Voyager output. It builds a decimated activity envelope, finds large “drive” spikes (median + $k \cdot 1.4826 \cdot \text{MAD}$), then in the trailing interval looks for delayed secondary peaks (*echo_z*), excess persistence, and a drive → echo spectral-centroid shift. Memory stays bounded regardless of trail length.

The README deserves explicit credit. It states plainly that a candidate is an anomaly that survives a noise null, not a foam detection; it lists the mundane sources that must be excluded (reaction wheels, heaters, ion-cyclotron, whistler, chorus); and it places the inferential burden on a pre-registered, confounder-controlled follow-up. The honesty of the framing is not in question here — only what the numbers turned out to be.

2. What was actually run, and what it produced

Both provided result files come from the transient lens; the steady-state lens output (foam_features.csv) was not included, so the line-search results are not assessed here.

Output file	Result
echo_events.csv (inner heliosphere: PSP / Solar Orbiter / MESSENGER / Helios)	Header only — 0 events detected.
voyager_echo_events.csv (Voyager tree)	117,370 drive-spike rows; 105,996 flagged candidate=1 (90.3%).

The inner-heliosphere zero is itself informative: with default gates (`--min-spike-snr 3 --min-echo-z 6 --min-persistence 3 --k 6`), those archives yielded no drive spikes that cleared the bar. All of the signal — and all of the trouble — is in the Voyager file.

3. The Voyager output in numbers

Quantity	Value
Drive-spike rows	117,370
Flagged candidate = 1	105,996 (90.3%)
Date range of spikes	1979-02-28 → 1980-11-16
Distinct source files	1,147
Reason: echo ($\pm$ downshift)	105,770
Reason: weak_spike / no_echo / persist($\times N$)	8,056 / 3,318 / a few hundred

Two features of this table do not survive contact with the underlying values: the 90.3% candidate rate, and the “echo (+downshift)” reason that accounts for almost all of it. The next two sections show why each is an artifact rather than a result.

4. Artifact one: echo_z is a divide-by-near-zero blow-up

The transient lens scores each candidate echo with a robust z against the trailing-window baseline. In `detect_echo`, that scale is computed as:

```
base = median(seg); mad = median(|seg - base|) + 1e-30
z = (seg[peak] - base) / (1.4826 * mad)
```

The $+ 1e-30$ floor prevents a divide-by-zero crash, but not a divide-by-*almost*-zero explosion. Whenever the trailing envelope is flat or quantized — long runs of identical or zero values, which is exactly what averaged, channelized, or gap-filled archive data looks like — the median absolute deviation collapses toward that $1e-30$ floor, and any tiny excess yields an astronomically large z . The data bear this out exactly: **echo_z reaches 3.16×10^{36} , and 41% of all rows carry $z > 10^{20}$** — values that cannot represent statistical significance. Because the candidate gate only asks for $z > 6$, essentially every one of these artifacts passes, which is what drives the 90% flag rate. The `candidate_reason` strings make it visible in plain text, e.g.

```
echo(z=337245379738297548249145278464.0, dt=48s)+downshift
```

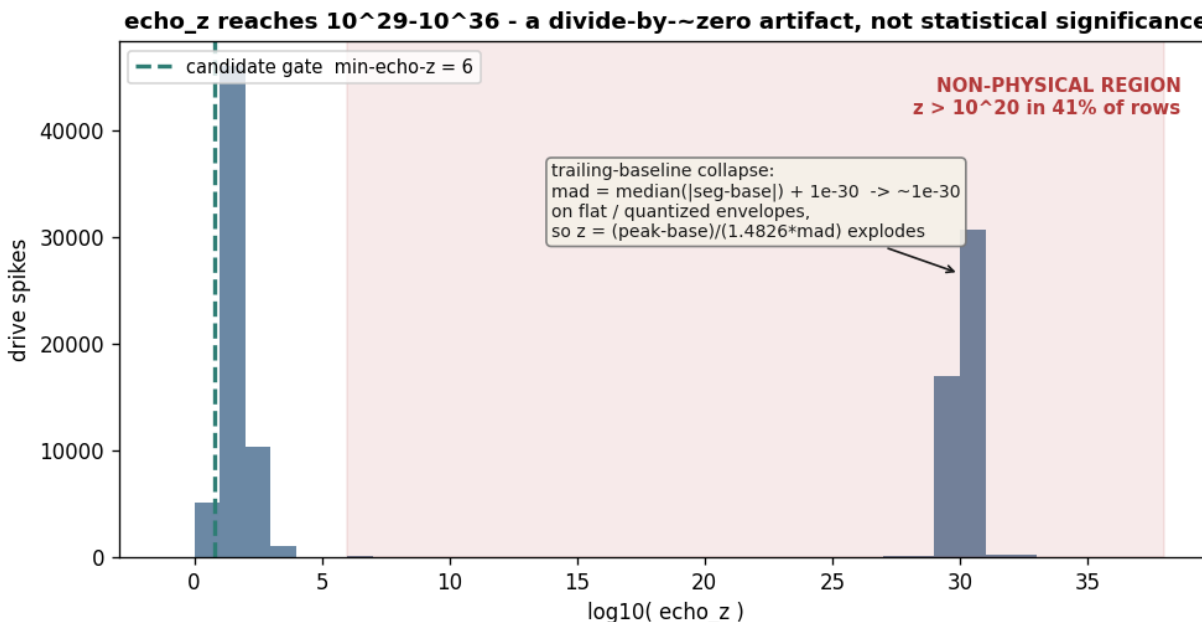


Figure 1. Distribution of $\log_{10}(\text{echo}_z)$. A finite cluster near $z \approx 10^{-10}$ sits beside a second population at 10^{29} – 10^{36} . The latter is non-physical and arises from the $\text{MAD} \rightarrow 0$ baseline collapse; it clears the min-echo-z = 6 gate trivially.

The same construction appears in spike detection (`detect_spikes` uses $\text{median} + k \cdot 1.4826 \cdot \text{MAD}$), which is why *spike_snr* likewise runs up to 3.9×10^{13} . These are not strong detections; they are the same arithmetic singularity seen from a different column.

5. Artifact two: the “downshift” is the negative half of zero

The signature borrowed from the Earth result is a *downward* frequency shift from drive to echo, and the pipeline tags any spike with `freq_shift < 0` as “+downshift.” But across the 111,068 rows that carry a shift, the distribution is centered on zero: **median +0.0012 Hz, mean +0.015 Hz, with only 36% of echoes shifting down** and nearly two-thirds shifting up. A genuine systematic redshift would move the whole distribution to the left of zero; instead the “downshift” tag is simply selecting the minority negative half of a symmetric scatter — a coin-flip dressed as an effect. (It was confirmed that the tag fires in exactly the rows with `freq_shift < 0`, and in no others.)

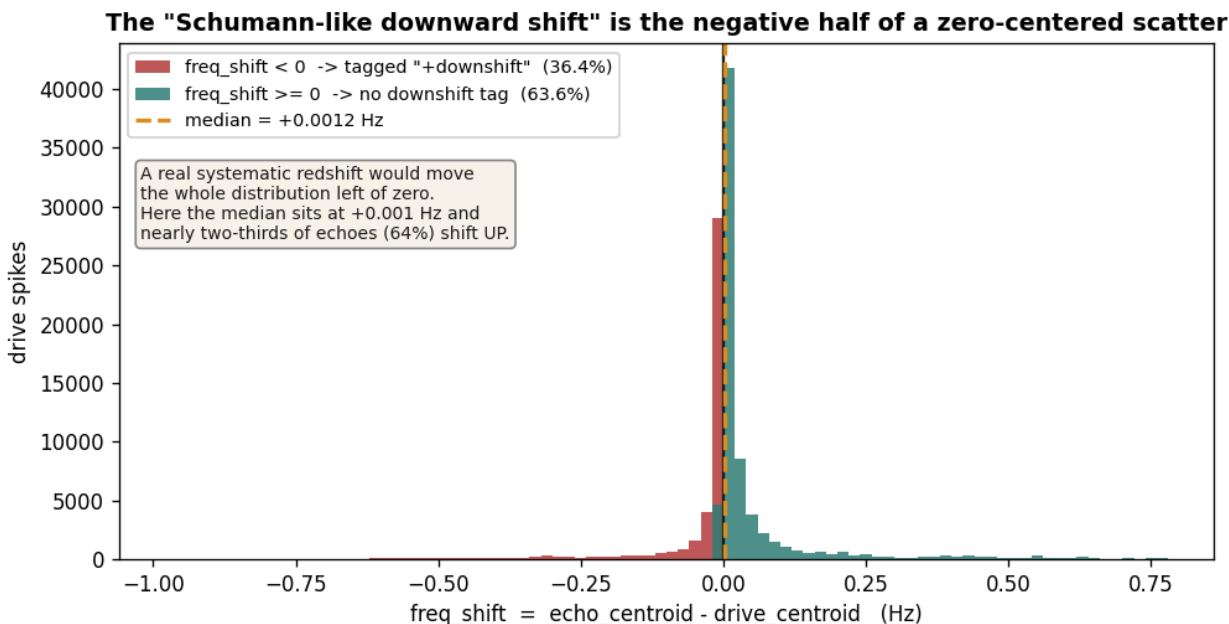


Figure 2. drive → echo frequency shift. The distribution is sharply peaked at zero (median +0.001 Hz). The red "downshift" population is the smaller side; more echoes shift up than down. There is no systematic Schumann-like redshift in these data.

6. What was actually measured

A third issue is what the detector was run on. Despite a Voyager tree containing thousands of Plasma Wave Subsystem (PWS) spectrum-analyzer files, **not one of the 117,370 rows comes from a PWS waveform file**. Every row originates from a plain .csv file — the large V179...CSV magnetometer/particle families and ELE_MOM.csv — read through the generic time-plus-numeric-columns path. The variable names are concatenations of whatever columns were found (STEP+MOV. +EG07, Electron_Density+Electron_Temperature), and **centroid_units is "Hz" for all 117,370 rows**, meaning everything was treated as a one-dimensional time series rather than a 2-D spectral product. In other words, the detector was applied to columns of unverified physical meaning, as if they were field magnitudes. Fill values were also not fully screened: **step_mean spans $\pm 9.99 \times 10^{10}$** , i.e. the Voyager fill sentinel ($\sim 10^{11}$) is leaking into the statistics.

7. Interpretation

Read honestly, this is a null — and a useful one. The inner-heliosphere echo search found nothing. The Voyager echo search produced six figures of "candidates" that are dominated by a baseline-collapse singularity (echo_z up to 10^{36}) and a frequency-shift tag that is centered on zero. Neither is evidence of a foam echo, and nothing here supports the single positive Earth result (the June 2026 Schumann triad) generalizing to spacecraft data.

The deeper lesson is methodological, and the README anticipated it: a permissive transient detector run across heterogeneous, quantized, gap-laden archive data will manufacture tens of thousands of "candidates" that are processing artifacts rather than physics. It is also worth naming an asymmetry inside the pipeline itself — the *steady-state* lens carries a rigorous parametric null (the Gamma/Bonferroni test), whereas the *transient* lens's z is a robust-baseline ratio with no such floor,

and it is the transient lens that generated these counts. The candidate column in this file should not be read as 105,996 interesting events; it should be read as a detector reporting on its own degenerate denominator.

8. What would make this a real test

None of this condemns the approach — the same tooling, with a few targeted fixes, could produce a defensible result. In rough order of leverage:

Floor the robust scale. Replace $\text{mad} = \text{median}(|\text{seg-base}|) + 1\text{e-}30$ with a floor tied to the signal itself — e.g. require a minimum number of distinct values in the trailing window, or set $\text{mad} = \max(\text{median}(|\text{seg-base}|, \epsilon \cdot \text{median}(\text{seg})))$ — and reject (rather than score) any window whose dispersion is near zero. This single change removes most of the 90% flag rate.

Make “downshift” a real test. Treat the frequency shift two-sided, pre-register an effect size, and compare against the $\approx 50\%$ that chance already delivers. A shift centered on zero is the null, not the signal.

Analyze verified channels only. Restrict to confirmed field magnitudes, and route PWS spectrum-analyzer data through the proper 2-D path with per-channel frequencies from the instrument table — the README already flags channel-index units, but this run went through the generic CSV path and labeled everything Hz.

Screen fill before enveloping. Reject $|\text{value}| \geq 10^{10}$ (and documented sentinels) before building the activity envelope, so the fill value stops entering the statistics.

Pre-register one known event. Run a single well-studied flyby — e.g. a Voyager Jupiter or Saturn bow-shock crossing — with a matched quiet-time control and a Kp-style confounder regression, exactly as the README’s “start with one or two well-studied events” advises, before any archive-wide scan.

Bottom line

The engineering and the honesty are real; the result is null. The inner-heliosphere echo run is a clean zero. The Voyager echo run is a null masquerading as 105,996 candidates — dominated by a divide-by-near-zero `echo_z` artifact (values to 10^{36}) and a “downshift” tag that is merely the negative half of a zero-centered distribution, computed on CSV columns of unverified meaning with fill values unscreened. Nothing supports a foam signature. With a robust-scale floor, proper PWS handling, fill screening, and a pre-registered single-event protocol, the same pipeline could be turned into a test whose candidates would actually mean something.